Cryptocurrency Market Prediction

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DS 5220 Final Report

Abstract

This project aims to provide a solution for a Kaggle Competition presented by DRW to predict the cryptocurrency market. This paper attempts to produce a model which can generate predictions with the highest Pearson Correlation with the true values of a hidden test set. We explore considerations on when to use complex models, why they may not always work, and when to use simple models in their place. We explore how regression based models can predict on unseen data and contribute to better understanding the cryptocurrency market in terms of price movement. We find that Lasso works the best in predicting on high dimensional minute-by-minute data and discuss its implications as well as limitations. Our findings are in line with other research in this new field of research in cryptocurrency and find that regression is a simple and powerful tool in forecasting.

Code — submitted with assignment

Datasets — https://www.kaggle.com/competitions/drw-crypto-market-prediction/overview

Introduction

In this project, I will be forecasting future crypto price movements based on minute-level historical data. The goal of the project is to develop a supervised machine learning algorithm which can accurately predict price movements in the cryptocurrency market for the hedge fund DRW. The problem inputs are train and test files with historical data representing proprietary features as well as price data regarding the market. The test file is only used to write the submission on Kaggle since it contains no target labels and the timestamps are shuffled, so for this project we will be working on the train file with labels and in-order timestamps. The train file contains mostly anonymous information about a cryptocurrency with features which represent economic indicators for the currency. There are 895 such features per timestamp as well as a target label which the model will try to estimate:

- **timestamp**: The timestamp index representing the minute associated with each row.
- **bid_qty**: The total quantity buyers are willing to purchase at the best (highest) bid price at the given timestamp.
- **ask_qty**: The total quantity sellers are offering to sell at the best (lowest) ask price at the given timestamp.
- **buy_qty**: The total trading quantity executed at the best ask price during the given minute.
- **sell_qty**: The total trading quantity executed at the best bid price during the given minute.
- volume: The total traded volume during the minute.
- **X_{1,...,780}**: A set of anonymized market features derived from proprietary data sources.
- **label**: The target variable representing the anonymized market price movement to be predicted.

There are 525,887 timestamps for this cryptocurrency representing different trading times on a minute basis ranging from 03/01/2023—02/29/2024. The evaluation of submissions to the competition is to based on achieving the highest Pearson Correlation coefficient between the predictions and the true labels on unseen test data in private testing conducted by DRW. The final output is a .csv file which contains the predictions on the test data and will be compared privately by the sponsor of the event with real market data at the end of the competition. This problem is compelling because cryptocurrency is so volatile and changes depending on often irrational behavior when compared with stocks. Since there is no tangible store of value, cryptocurrency prices are a direct reflection of what people perceive the value to be at a given moment, so the prices fluctuate greatly and unpredictably with bonds. Conversely, stocks are backed by real assets which are ultimately backed by the bonds of a particular government. The extreme volatility and novelty of cryptocurrency means there is a great deal to discover about its behavior and different models may be useful in predicting cryptocurrency when compared to predicting stocks.

Proposed Solutions

The Kaggle Competition requires the final model to work well on unseen data, avoid overfitting, and may not future peek during training. This means that an ordinary ARIMA

model will not be valid for the competition, since the timestamps on the test data are randomized and masked. However, we can use a simple rolling mean and ARIMA model to form a baseline for comparison to more complex models. Typically, for highly dimensional data, Neural Networks, Decision Trees, Random Forests, and Hidden Markov chains work well to predict on unseen data. For this project, these models did not work any better than random and were prone to overfitting or never converging in the case of neural networks. The most promising model in early testing was the Ridge model which outperformed Linear Regression. This lead me to explore further regression models which could work.

Background

This project is produced in python using algorithms from the Sklearn library. The data comes in parquet file format and requires to be read in using pandas to be stored in a dataframe. Note for evaluation we have reserved 20% of the train data for final testing. We will be using the Pearson Correlation Formula to cross validate in the training, for evaluation on the final testing set and to generate predictions in line with the competition expectations. The project will be testing different regression algorithms such as Ridge, Lasso, and ElasticNet to evaluate which algorithms perform the best on this highly dimensional, anonymous, and highly random data.

Algorithms

The Pearson Correlation Formula is used in cross validation within folds of the data to select the Mean Squared Error (MSE) with the highest correlation in each iteration of training.

$$\rho_{X,Y} = \frac{\mathbb{E}[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$$

Figure 1: Pearson Correlation Formula (Wikipedia)

We test the Ridge algorithm which performs as Linear Regression but with a regularization term of L2 which has the effect of applying a linear penalty to the weights. This results in the weights shrinking to near zero but never quite reaching 0. This model cannot perform feature selection and may be sensitive to noise with such high dimensional data.

$$\widehat{\beta}\,(\mathsf{ridge}) = \underset{\beta}{\operatorname{argmin}} \| \mathbf{y} - \mathbf{X}\beta \|_2^2 + \lambda \| \beta \|_2^2$$

Figure 2: Ridge Algorithm (Ogutu 2012)

Lasso works similar to Ridge in the sense of adding a regularization term to ordinary Linear Regression, but goes a step further be applying L1 regularization instead of L2. This results in the ability to shrink coefficients down to zero and effectively perform feature selection by selecting only predictive features in the data, disregarding other features.

$$\widehat{\beta}\,(lasso) = \underset{\beta}{\operatorname{argmin}} \| \mathbf{y} - \mathbf{X}\beta \|_2^2 + \lambda \| \beta \|_1$$

Figure 3: Lasso Algorithm (Ogutu 2012)

ElasticNet is another promising algorithm as it combines the power of L1 and L2 regularization into a single algorithm. It is capable of automatic variable selection and lowering weights linearly to non-zero values. This flexibility allows this algorithm to find correlated features and increase their weights while lowering the weights of less correlated features.

$$\widehat{\beta}\,(\mathrm{enet}) = \left(1 + \frac{\lambda_2}{n}\right) \left(\left\{ \underset{\beta}{\operatorname{argmin}} \| \mathbf{y}\text{-}\mathbf{X}\beta \|_2^2 + \lambda_2 \| \beta \|_2^2 + \lambda_1 \| \beta \|_1 \right) \right\}$$

Figure 4: ElasticNet Algorithm (Ogutu 2012)

In training, we modified the training scorer to find the MSE which produces the highest Pearson Correlation, so the models are not penalized directly by MSE, rather by the Pearson Score. Since this is the evaluation metric by the competition, this is a good metric to penalize the models with.

Related Work

In the field of predicting cryptocurrency this paper (Ashok 2025) explores the use of Gradient Boosting or Random Forests as a potential method for predicting cryptocurrency price changes for the major cryptocurrencies. The study suggests this is a difficult task to apply more standard time series approaches and approaches the problem and shows the potential to approach cryptocurrency price as a problem for various tree type algorithms.

Alternatively, (Qureshi 2025) confirms that classifier model based on decision trees such as XGBoost, random forests, and gradient boosting are produce the best results when predicting price changes in crypto. When comparing the various methodologies there are trade offs between short term and long term predictive power. Neural networks appear to work better for long term predictions whereas for high frequency trading the classifiers appear to perform better.

Another approach states that logistic regression is the most powerful model in market forecasting for cryptocurrency (Islam 2025). Given the decentralized nature of cryptocurrency it allows for many manipulative tactics to be employed in regular trading such as pump and dump, wash trading, and spoofing. These tactics pollute the data and make it more difficult for a machine learning model to capture the patterns underlying the data. In testing, the benchmark for a decent model is if it exceeds 50% accuracy in testing as a classifier. Given the extreme volatility of cryptocurrency, market prediction in this domain is very much still an unsolved problem.

Finally, the most complex approach is the use of transformers (Mahdi 2025). This paper employs sentiment analysis as a signal for trading along with historical price data, volume, and the Fear and Greed Index. The results are promising showing much less error than other models. However, the results on actual trading are unclear, nevertheless transformers are very complex and heavyweight implementations which are beyond the scope of using a simpler model to predict cryptocurrency.

Project Description

Initially we define the functions for the Pearson correlation computation and the custom scorer for training. Additionally to reduce memory, we convert integers and floats to smaller byte values for efficient memory since the file is quite large. We then engineered some features such as:

- **order_imbalance**: (bid - ask) / (bid + ask + norm) Measures interest in buying
- **trade_imbalance**: (buy - sell) / (buy + sell + norm) Measures buy trades vs. sell trades
- **buy_sell_ratio_log**: np.log((buy + norm)/ (sell + norm)) Measures the ratio of buy/sell in a log scale for normalization
- **volume_ratio**: volume / (buy + sell + norm) Accounts for trading volume not accounted for by the best bid and ask
- **participation**: (buy + sell) / (volume + norm) Measures the ratio of trades that happen at the best bid or ask

Preliminary Testing

According with existing literature, similar problems with predicting cryptocurrency behavior have been solved to some degree with models such as decision trees, random forests, and neural networks. Given our data is highly dimensional, we tested the performance of these algorithms but failed to observe nearly any correlation with the unseen test set. In early testing, the Ridge Algorithm showed some predictive potential and was much less time and memory intensive to run. In private testing on unseen data the Kaggle leaderboard shows the best solution as receiving a Pearson score of 0.13959. Initial results with Ridge showed results ranging from 0.04—0.08 correlation whereas with all other models there was no correlation score higher than 0.01. Therefore for the project, we implemented the three algorithms Ridge, Lasso, and ElasticNet and performed grid search to obtain the optimal hyper-parameters for each. Then results were compared in terms of highest Pearson Correlation.

Baseline Model

The baseline ARIMA and rolling average models are not valid for the submissions since they require time sensitive knowledge, and for the competition there must be no future peeking polluting the training data. The ARIMA model does not perform well on this data, and the Rolling Average works well but is not valid for the submission.

Final Model

The baseline ARIMA and rolling average are useful to compare with the regression models we use in terms of ensuring that we are working better than random. In our final model, we are re-reading the files and ensuring we have a time-sensitive split for cross-validation. Additionally we are dropping missing values and imputing them with SimpleImputer before scaling with StandardScaler within the pipeline. Then we perform a grid search in order to find the best hyper-parameters per model which can produce the best Pearson correlation r score on the holdout test data. We have also accounted for repeating rows, infinite values, and rows where all values may be the same. During cross validation, we are doing four splits and within each fold we test on a fraction of the data in order to update the weights for the next fold until finally we have an out of fold prediction on the test set. I then plot the results along with the true labels. The strongest final model is Lasso, followed by ElasticNet and finally Ridge. Lasso performs the best because it us able to use L1 regularization in order to drive weights of less useful features down to zero.

Empirical Results

We start by testing the rolling mean, and for this we will separate the data in such a way that we have future-peeking just to demonstrate a baseline model. In the initial ARIMA and rolling averages model, we have a Pearson r score of 0.3475 (Figure 5) for the rolling average model and a score of 0.0091 (Figure 6) for the ARIMA model. Overall, these scores are not strong, especially given future peeking.

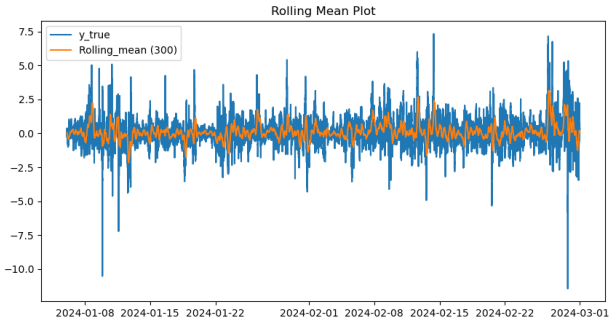


Figure 5: Rolling Mean plot, r_score = 0.3475

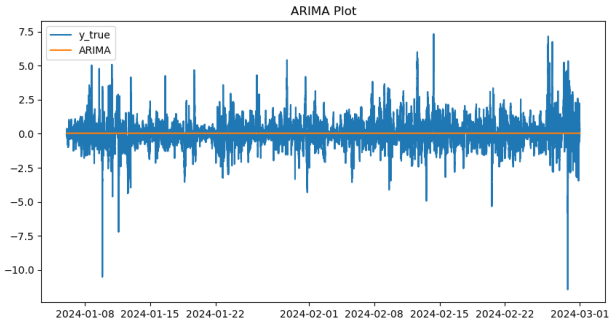


Figure 6: ARIMA (2, 1, 1) plot, r_score = 0.0091

In order to improve the scores past baseline, we employ three models and train them on the new processed data. In each fold of cross validation we are updating the weights based on the highest achieved Pearson score until we are finally predicting on the holdout test data. In the final predictions we can measure the ROC curve for the threshold of target > 0 to see how well the model classifies positive movement. In this chart (Figure 7), all models appear to be performing well and similar in terms of capturing the positive movement.

Plotting the actual values vs the predicted values for each chart highlights the main difference between the Ridge model and the other two models. Ridge has a massive negative prediction which the other two lack, and also does not follow trends as closely as ElasticNet or Lasso. (Figure 8)

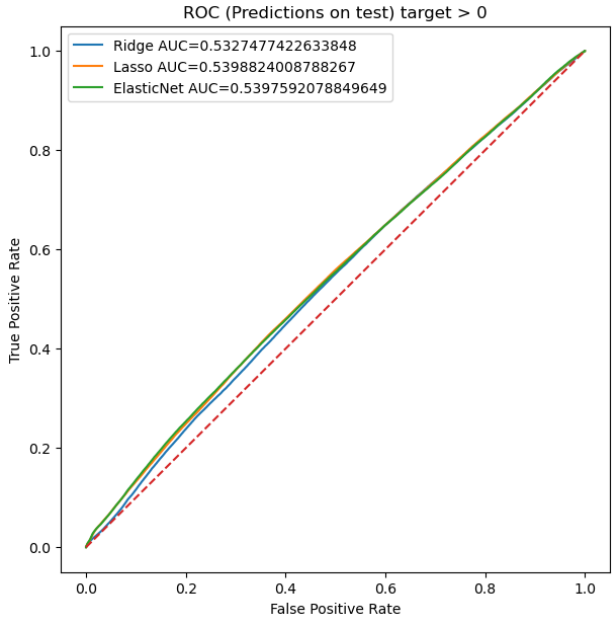


Figure 7: ROC plot for Ridge, Lasso, and ElasticNet

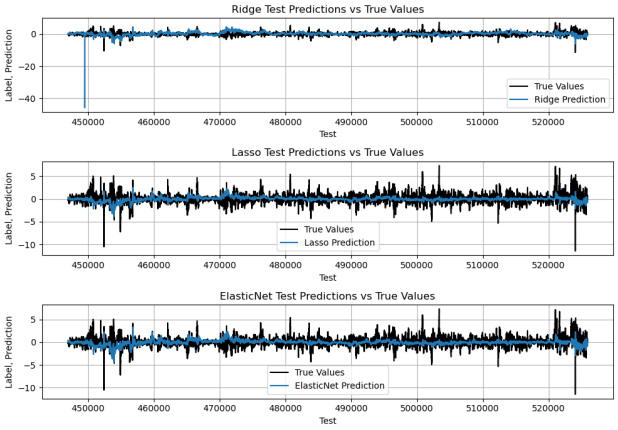


Figure 8: Ridge, Lasso, and ElasticNet predicted vs. true values plot

In terms of performance, Lasso and ElasticNet have only minor predictive differences but respond more reasonably to market noise.

Scores:

- **Lasso r score**: 0.1288
- **ElasticNet r score**: 0.1209
- **Ridge r score**: 0.0446

We can infer that Lasso is the best performer among the regressors because it is able to bring weights down to zero

for features it deems not important; that is the only difference between Lasso and the other algorithms (ElasticNet can do this but in practice is not doing it enough). Overall, regression works best when applied to this dataset in terms of producing the highest r score compared to the true labels. We perform better than a baseline ARIMA model which has future peeking in its training set, but not better than the rolling average (which cannot be used for this type of financial data, it merely serves as another baseline metric). In final analysis Lasso has a slight edge over the other models but is not a great performer in terms of capturing a very high correlation with the true values.

Conclusions

In terms of accurately predicting the anonymous target, which most likely corresponds to some type of price change index, our model does not perform much better than random guessing. In terms of other submissions for the competition, the metrics of our top model perform only about 10-15% worse than the top scorer on our own testing on holdout data. Given the nature of the problem, that it is difficult and unsolved, our results may be in fact promising.

Implications

Our findings are in line with several other findings in this field; that forms of regression are powerful predictors of the cryptocurrency market. When compared with more complex models regression simply outperforms on real test data. This is likely because regression does not overfit by nature and is less sensitive to noise. In this data we clearly had far too much noise and many of the provided indicators turned out useless if not harmful to model performance. Lasso worked the best because it was able to perform feature selection and to bring certain features down to zero weight. We performed a grid search to find the alpha which would deliver the best results in terms of the bias variance trade off. ElasticNet was close behind in terms of performance but did not perform as well because it may have kept certain features which Lasso made zero weight. Since we have 895 predictors to start (a large number) if even one of those predictors is a large noise, it can throw off the entire prediction. Ridge regression is unable to perform feature selection and therefore did not perform as strongly as the others, but still outperformed any other non-regression based model by far.

Future Work

The anonymous nature of this data makes it difficult if not impossible to add additional data for predictions, but if possible this would be the next logical step in improving performance. It seems many of the provided features were very weakly correlated to the target label. This work can be expanded through more feature engineering and through insider knowledge on what metrics were included in the raw data. If I had more time on the project I would try to engineer more features and to exclude some features entirely from the data as many of them appear to be noisy. Then I would implement the transformer based architecture presented earlier (Mahdi 2025) and include other data sources as the paper did. Since this is a competition and the client does not want to disclose information about what the project is really for, it is difficult for me to provide suggestions on how to improve the work given this data was all that was provided.

Advice

For future students I suggest that a general simple model is better than an overly complex model. In real life, simple linear regression can actually model macro trends very well. Complex neural networks, decision trees, and random forests do not always work better in practice because they are still sensitive to overfitting, never converging, and may not generalize in practice. If there are many features, try to determine collinearity or use a model that has feature discrimination embedded in the algorithm, such as Lasso. Reducing dimensionality of the data can allow the real signals in the data to become more prominent.

References

Kaggle Competition
https://www.kaggle.com/competitions/drw-crypto-market-prediction/overview

Pearson Correlation Coefficient
https://en.wikipedia.org/wiki/Pearson_correlation_coefficient

Genomic selection using regularized linear regression models: ridge regression, lasso, elastic net and their extensions
Ogutu, Joseph O et al. BMC proceedings vol. 6 Suppl 2,Suppl 2 (2012): S10. doi:10.1186/1753-6561-6-S2-S10

Comparing Machine Learning Algorithms for Predicting Cryptocurrency Prices: An Analytical Study.
Ashok, Pappula. (2025). Advances in Nonlinear Variational Inequalities. 28. 472-482. 10.52783/anvi.v28.3504.

Evaluating machine learning models for predictive accuracy in cryptocurrency price forecasting
Qureshi SM, Saeed A, Ahmad F, Khattak AR, Almotiri SH, Al Ghamdi MA, Shah Rukh M. 2025. PeerJ Computer Science 11:e2626 https://doi.org/10.7717/peerj-cs.2626

Machine Learning-Based Cryptocurrency Prediction: Enhancing Market Forecasting With Advanced Predictive Models
Islam, M. S., M. Bashir, S. . Rahman, M. A. . Al Montaser, J. C. . Bortty, A. . Nishan, and M. R. .Haque. Journal of Ecohumanism, vol. 4, no. 2, Mar. 2025, pp. 2498 -, doi:10.62754/joe.v4i2.6663.

A Novel Hybrid Approach Using an Attention-Based Transformer + GRU Model for Predicting Cryptocurrency Prices.

Mahdi E, Martin-Barreiro C, Cabezas X. Mathematics. 2025; 13(9):1484. https://doi.org/10.3390/math13091484